

Properties of the Z-Transform

	Property	Discrete Time Domain	$\mathcal{Z}$ Transform
1	Linearity	$af_1[n] + bf_2[n] + \dots$	$aF_1(z) + bF_2(z) + \dots$
2	Shift of $x[n]u_0[n]$	$f[n - m]u_0[n - m]$	$z^{-m}F(z)$
3	Left shift	$f[n - m]$	$z^{-m}F(z) + \sum_{n=0}^{m-1} f[n - m]z^{-n}$
4	Right shift	$f[n + m]$	$z^m F(z) + \sum_{n=-m}^{-1} f[n + m]z^{-n}$
5	Multiplication by a^n	$a^n f[n]$	$F\left(\frac{z}{a}\right)$
6	Multiplication by e^{-nsT_s}	$e^{-nsT_s} f[n]$	$F(e^{sT_s} z)$
7	Multiplication by n	$nf[n]$	$-z \frac{d}{dz} F(z)$
8	Multiplication by n^2	$n^2 f[n]$	$-z \frac{d}{dz} F(z) + z^2 \frac{d^2}{dz^2} F(z)$
9	Summation in time	$\sum_{m=0}^n f[m]$	$\frac{z}{z-1} F(z)$
10	Time convolution	$f_1[n] * f_2[n]$	$F_1(z)F_2(z)$
11	Frequency convolution	$f_1[n]f_2[n]$	$\frac{1}{j2\pi} \oint x F_1(v) F_2\left(\frac{z}{v}\right) v^{-1} dv$
12	Initial value theorem	$f[0] = \lim_{z \rightarrow \infty} F(z)$	
13	Final value theorem	$\lim_{n \rightarrow \infty} f[n] = \lim_{z \rightarrow 1} (z-1)F(z)$	

See also: [Wikibooks: Engineering Tables/Z Transform Properties](#) and [Z-Transform—WolframMathworld](#) for more complete references.

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